SEID 2364

Assignment 1 – Moral Architecture, Aggregation, and Data Access
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Due: Week 2

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Purpose

This assignment establishes your baseline ethical position, introduces you to large-scale moral aggregation research, and begins your semester-long Ethical Journal (lab log).

You are not being asked for the "correct" answer. You are being asked to begin constructing a defensible analytical framework.

Systems Setup

Account Preparation (Required Before Next Class)

Even though you will not use GitHub for submission this week, you MUST create and verify the following accounts before the next class session:

1. GitHub

- Create a GitHub for Education account (use a professional username). **Link to Setup**
- Verify your email.
- Ensure you can log in from your laptop.

We will initialize repositories together in class.

2. Zotero

- Create a **Zotero** account. (Zotero.com)
- **Install Zotero** (desktop application AND browser connector).
- Join the shared SEID 2364 group library (details will be provided in class).

You should arrive next class with both accounts active and accessible.

Part I – Journal Setup (Temporary – Pre-GitHub)

You will NOT use GitHub for this first submission. We will formally set up repositories next week.

For now:

- Create a working document in a format you control (Google Doc, Word, Markdown, etc.).
- Title the document: > SEID 2364 – Ethical Journal
- Save it in a location where you have reliable access.

This document will evolve throughout the semester. Next week, you will transfer or link this journal into your GitHub repository.

Part II – Journal Entry 1

Title your document:

> Journal Entry 1 – Baseline Ethics and Moral Aggregation

Section A – Original Definition

Reproduce verbatim your in-class response:

> What does ethics mean to you?

Do not edit this section.

Section B – Revision and Analysis

After our Day 1 discussion, respond to the following:

1. Has your definition changed? Why?
2. Does ethics require more than one intelligence?
3. Do machines that interact with society require ethical architecture?
4. Are there universal ethical principles, or are ethical systems historically and culturally contingent?
5. Under what conditions would your position fail?

Be precise. Define key terms.

Part III – Moral Machine Experiment

A. Participate

Complete an extended session of the MIT Moral Machine experiment.

Document:

- Screenshot of your summary results.
- Observed preference patterns.
- Any demographic or clustering results provided.

B. Core Reading

You must read:

1. Takemoto (2024), *The Moral Machine experiment on large language models*, Royal Society Open Science. https://royalsocietypublishing.org/rsos/article/11/2/231393/92489/The-moral-machine-experiment-on-large-language

2. Takemoto (2023), *Data and Code on the Moral Machine experiment on LLMs* (Dryad dataset). https://doi.org/10.5061/DRYAD.D7WM37Q6V

C. Analytical Response

Add a section titled:

Moral Aggregation and Legitimacy

Respond:

1. Did your decisions follow a consistent rule?
2. Where did your rule break down?
3. Should majority preference determine autonomous system behavior?
4. Does aggregated data equal moral legitimacy?
5. What is being measured in the Moral Machine: morality, preference, or something else?

Part IV – Data Access, Code Inspection, and Computational Pipeline

You are data science majors. You are expected to engage directly with the computational structure.

Using the Dryad dataset and code archive:

https://doi.org/10.5061/DRYAD.D7WM37Q6V

A. Inspect the Repository Structure

Download the archive and examine:

- The scenario generation script (`generate_moral_machine_scenarios.py`)
- Model execution scripts (`run_chatgpt.py`, `run_palm2.py`, `run_llama2.py`)
- The CSV conversion script (`convert_pickle_csv.py`)
- The R scripts used for analysis

You are NOT required to run large experiments. You ARE required to understand the pipeline.

B. Pipeline Analysis (Written Response Required)

In your journal, explain:

1. How are Moral Machine scenarios generated computationally?
2. How are model responses collected?
3. How are outcomes encoded in the CSV files?
4. What variables define a "moral decision" in this dataset?
5. What assumptions are built into the scenario categories (Age, Gender, Social Status, Species, Utilitarian, etc.)?

C. Optional (Technical Stretch – Not Required)

If you have the environment available:

- Run a very small number of scenarios (e.g., 10–50).
- Inspect the resulting CSV structure.

You are NOT required to reproduce published figures. This is exploratory.

Part V – Encoding Ethics

Respond thoughtfully: this is NOT a research topic, but a speculative thought experiment.

1. Can you encode your current working definition of ethics into a machine? Why or why not?
2. If machines interact with society, what must ethical architecture include?
3. Are there ethical questions that may be structurally undecidable?
4. Can computational systems fully resolve moral ambiguity?

Submission and Preparation

You will NOT submit this assignment before next class.

Instead:

- Complete all components of Journal Entry 1.
- Bring your completed document (digital copy) to the next class session.
- Ensure you can access it from your laptop.
- Ensure all screenshots and materials are embedded or accessible.

In the next class, we will:

- Set up GitHub repositories.
- Demonstrate structured submission.
- Transfer or link your Ethical Journal into version control.

You must arrive prepared to submit.

Evaluation Criteria

You are evaluated on:

- Clarity of reasoning
- Precision of definitions
- Structural understanding of mediation and aggregation
- Ability to distinguish preference from legitimacy
- Recognition of computational and formal limits

Agreement with any particular moral stance is NOT evaluated.